Branch Specialization in Attention

Phase 1 Attention-Role Specialization Checkpoint

Research checkpoint

May 21, 2026

Current Question

Question: Do transformer heads have stable functional roles across seeds?

Phase 0 showed weak universality in raw attention-score matrices: same-index similarity was low, but Hungarian matching recovered corresponding heads.

Phase 1 now asks whether simple role proxies $S(h, t)$ show the same pattern.

Role Proxies

- **BOS**: attention mass to token position 0.
- **Previous-token**: attention mass from position i to $i-1$.
- **Repeat-match**: on synthetic repeated-token sequences $[x_1, \ldots, x_n, x_1, \ldots, x_n]$, attention from the second x_i to the first x_i.

Scores are normalized across heads within a layer:

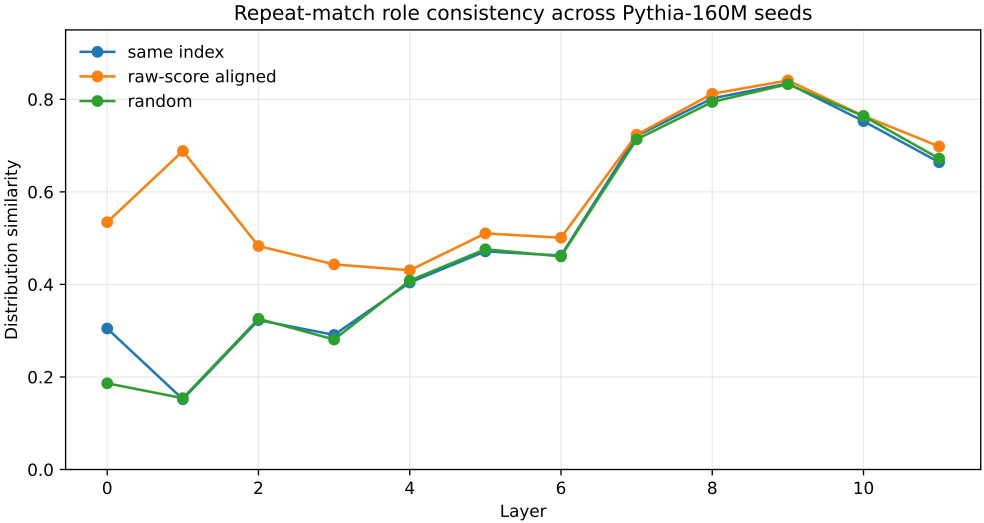
$$S(h,t) = \frac{\mathrm{score}(h,t)}{\sum_{h'} \mathrm{score}(h',t)}$$

Aggregate Cross-Seed Consistency

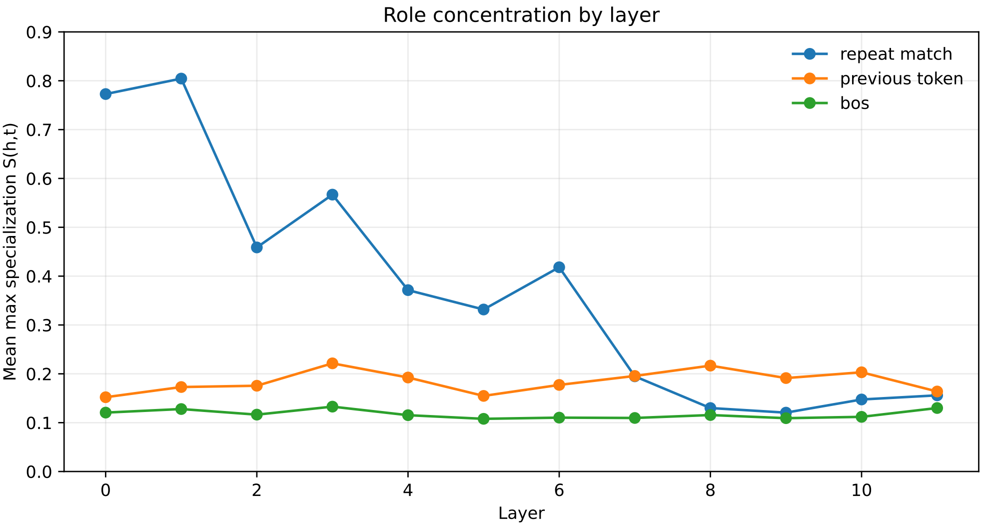
Role	Raw	Aligned	Random
BOS	0.8317	0.8566	0.8310
Previous-token	0.7531	0.8116	0.7531
Repeat-match	0.5152	0.6191	0.5057

BOS is diffuse. Repeat-match has the clearest role-specific aligned-over-random signal.

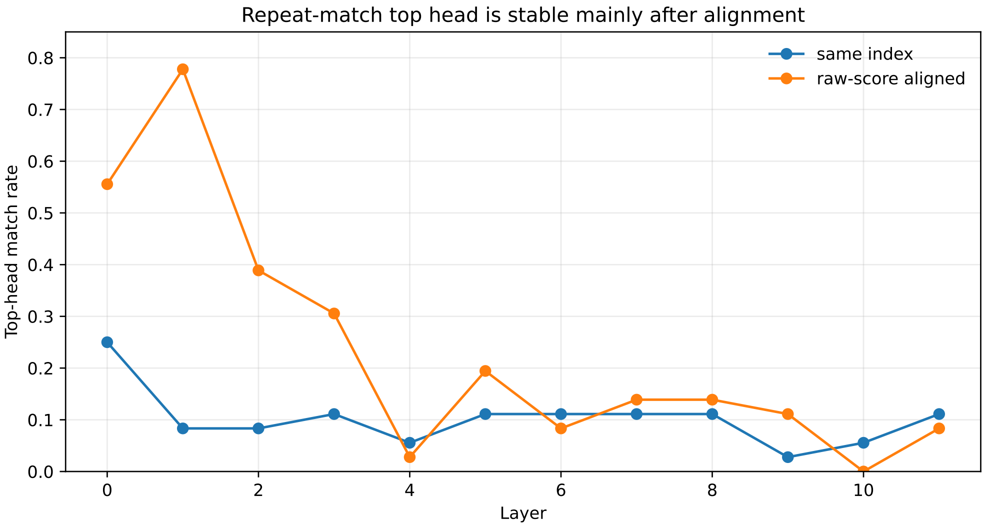
Repeat-Match Consistency



Role Concentration



Top-Head Stability



Key Finding

Repeat-match is highly concentrated in early layers:

Role	Layer	Mean max $S(h,t)$	Effective heads
Repeat-match	0	0.7728	2.84
Repeat-match	1	0.8045	2.26
Previous-token	3	0.2215	10.31
BOS	11	0.1300	10.60

This is the first clear branch-specialization-like signal in the project.

Interpretation

- A concentrated repeat-match attention role appears across Pythia-160M seeds.
- The responsible same-index head is unstable.
- Raw attention-score alignment recovers much stronger top-head agreement.
- This supports weak universality for a concrete role proxy, not just generic attention similarity.

Caveats

- This is still an attention-pattern proxy, not causal patching.
- Repeat-match is induction-style, but not a full induction-copying behavioral test.
- BOS is not useful here because it is distributed and near random under permutation.
- Next step must test whether these heads causally affect repeated-token prediction.

Decision

Continue with repeat-match / induction-style heads as the first causal target.

Next decisive experiment:

- identify top repeat-match heads in layers 0–1 for each seed;
- ablate those heads on repeated-token next-token prediction;
- compare against random heads and aligned corresponding heads.

Provenance

- Script: `scripts/attention_role_specialization.py`
- Output: `results/phase1_pythia160m_attention_role_specialization`
- Report: `doc/phase1_pythia160m_attention_role_specialization.md`
- Alignment input: `results/phase0_pythia160m_seed1_to_9_step143000_raw_scores/summary.json`
- Data copied into this deck folder under `data/`